

Open and Closed Balls

- The set $B(x, r) = \{y \in X : d(x, y) < r\}$ is called the open ball $B(x, r)$ with a center x and a radius r .
- The set $B(x, r) = \{y \in X : d(x, y) \leq r\}$ is called the closed ball $B(x, r)$ with center x and radius r .

A closed ball contains all the points of the boundary $d(x, y) = r$. An open ball does not contain these points. Sometimes the radius is labeled as ϵ instead of r and the ball is called an epsilon ball.

Balls in $\mathbb{R}^1$

- What is an open ball in $\mathbb{R}$?
 - An open ball in $\mathbb{R}$ is an open interval $(x - r, x + r)$, with x as the midpoint. Formally, it is the set $\{y \in \mathbb{R} : x - r < y < x + r\}$
 - A closed ball in $\mathbb{R}$ is a closed interval $[x - r, x + r]$, with x as the midpoint. Formally, it is the set $\{y \in \mathbb{R} : x - r \leq y \leq x + r\}$

Sequences

Definition A *sequence* is an assignment of the elements in some set to the natural numbers. A sequence is denoted as a set with elements labeled from zero (or one) to a finite number or infinity:

- Finite sequence: $\{x_n\}_{n=0}^N = \{x_1, x_2, \dots, x_N\}$
- Infinite sequence: $\{x_n\}_{n=0}^{\infty} = \{x_1, x_2, \dots\}$

Examples:

- $\{x_n\}_{n=0}^{\infty} = \{1, 1, 2, 3, 5, 8, \dots\}$
- $\{x_n\}_{n=0}^{\infty} = \{1, 0, 1, 0, \dots\}$
- $\{x_n\}_{n=0}^{\infty} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$

Sequences as functions of n

Sequences can also be defined as functions of n . For example,

- $a(n) = n^2 \implies \{x_n\}_{n=0}^{\infty} = \{0, 1, 4, 9, 16, \dots\}$
- $b(n) = \frac{n}{n+1} \implies \{x_n\}_{n=0}^{\infty} = \{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\}$
- $c(n) = \sqrt{\ln n} \implies \{x_n\}_{n=0}^{\infty} = \{\sqrt{\ln 1}, \sqrt{\ln 2}, \dots\}$
- $d(n) = \frac{e^n}{n} \implies \{x_n\}_{n=0}^{\infty} = \{e, \frac{e^2}{2}, \frac{e^3}{3}, \dots\}$

Series

A **Series** is a sum of a sequence. For example,

- $\sum_{i=1}^N x_i = 1 + 2 + 3 + \dots$
- A common series that you will see in game theory is a discount rate: $\sum_{t=0}^{\infty} \delta^t$
- How should a state value future consumption versus current consumption?

Calculating Discounted Utility

Say that the state's GDP for year i is defined as x_i . Then we can model the value of all future GDP as $\sum_{t=0}^{\infty} x_i \delta^t$. How do we calculate this value?

- Define the present value as $S = \sum_{t=0}^{\infty} x_i \delta^t = x_i(1 + \delta + \delta^2 + \dots)$
- Multiply both sides by δ to get $\delta S = \sum_{t=1}^{\infty} x_i \delta^t = x_i(\delta + \delta^2 + \dots)$
- Subtract the second equation from the first equation:
$$(1 - \delta)S = x_i$$
- Divide both sides by $(1 - \delta)$ to get $S = \frac{x_i}{1 - \delta}$

Limits, Convergence, and Divergence

A Limit of a Sequence $\{x_i\}$ is a number L such that $\lim_{i \rightarrow \infty} x_i = L$.

Definition A sequence $\{x_n\}_{n=1}^{\infty}$ in $\mathbb{R}$ has a *limit* $L \in \mathbb{R}$ iff for each $\epsilon > 0$, $\exists K \in \mathbb{Z}^{++}$ such that if $n \geq K$, then $|x_n - L| < \epsilon$. If L exists, write $\{x_n\}_{n=1}^{\infty} \rightarrow L$ or $\lim_{n \rightarrow \infty} x_n = L$.

The elements in the sequence get arbitrarily close to L as the index i gets larger.

- There must be a number K such that all elements after the K th element must be in the epsilon ball $B_{\epsilon}(L)$.
- A sequence *converges* if it has a limit. If it has no limit (or if it has a limit of $\pm\infty$) then it *diverges*.

Examples Convergent and Divergent Sequences

- $\{i\}_{i=1}^{\infty}$
- $\{\frac{3}{10^i}\}_{i=1}^{\infty}$
- What is $\lim_{i \rightarrow \infty} \delta^i$ if $|\delta| < 1$?
- What is $\lim_{i \rightarrow \infty} \frac{1}{i^z}$ if $z > 0$?

Limits of Series

If the series is $\sum_{i=1}^N x_i$, then the limit is $\lim_{N \rightarrow \infty} \sum_{i=1}^N x_i = S$.
For example, $\lim_{N \rightarrow \infty} \sum_{t=0}^N \delta^t = \frac{1}{1-\delta}$.

Limits of Functions

Definition: $\lim_{x \rightarrow p} f(x) = q$ if for every $\epsilon > 0$ we can find $\delta > 0$ such that $|f(x) - q| < \epsilon$ for all x for which $|x - p| < \delta$. q is called the limit of f at the point p .

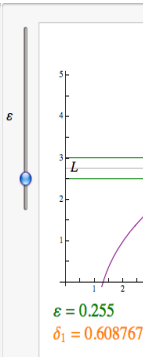
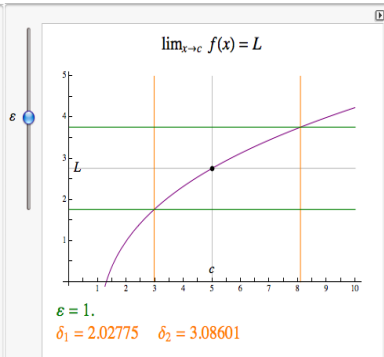
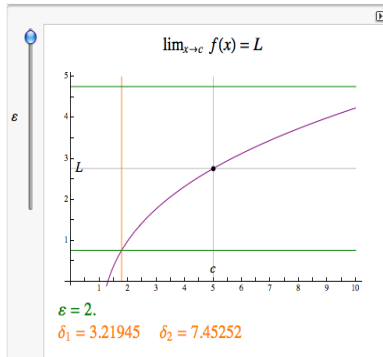
If $\lim_{x \rightarrow p} f(x) = q$ then $f(x)$ becomes arbitrarily close to q given arbitrarily small movements towards p .

- If x approaches p from above, we write $\lim_{x \rightarrow p^+} f(x) = L^+$
- If x approaches p from below, we write $\lim_{x \rightarrow p^-} f(x) = L^-$
- If $L^+ = L^-$, then the function has a unique limit at p

Graphical Illustration of Limits

$$\lim_{x \rightarrow c} f(x) = L$$

if for every $\epsilon > 0$ we can find $\delta > 0$ such that $|f(x) - L| < \epsilon$ for all x for which $|x - c| < \delta$. L is called the limit of f at the point c .



Case when $f(x) \neq L$

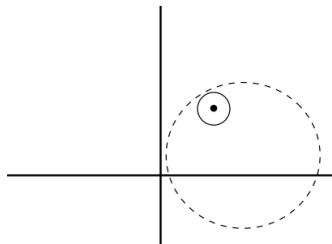
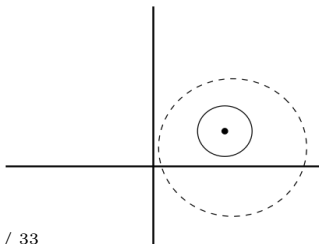
The existence of a limit depends upon the behavior of the function near a specific value of a variable(s) defined in the function, which is not the same as the value of the function at the specific

variables. For example: $f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ (x - 2)^2 & \text{if } x \geq 2 \end{cases}$

See page 91 for more properties on limits.

Open Sets

- **Definition:** A set U is *open* if $\forall x \in U$ there exists $r > 0$ such that $B(x, r) \subset U$.
- For an open set there is some distance that you can move in any direction and remain within the set.
 - This means that a set is open if for any point in the set we can find a small ball around x that is also contained within the set.
 - What would a set need to contain to violate this definition?



Examples of Open Sets

- Is $(0, 1)$ an open set?
- Is $[5, 10]$ an open set?
- Is ∞ an open set?

Closed Sets

- A set U is *closed* if its complement is open.
 - A closed set is a set that contains its boundary, which makes its complement open.
 - For example, the interval $[0, 1]$ is closed, since its complement set $(-\infty, 0) \cup (1, \infty)$ is open.
- An equivalent definition is that a set is closed iff $\forall$ sequences $\{x_k\}$ with $x_k \in U \ \forall k$ and $\{x_k\} \rightarrow x$, then $x \in U$.
 - If a sequence $\{x_k\}$ gets arbitrarily close to a point x while staying in the closed set then the set must also contain x .
 - For example, the sequence $\frac{1}{n}$ converges to 0 and all elements in this sequence are contained within the interval of 0 to 1. The open interval $(0, 1)$ violates this definition, while the closed interval $[0, 1]$ does not violate the definition of a closed set for this particular sequence.

Closed Sets Continued

- Finally, note that sets do not have to be open or closed. What is an example?
- Can a set be both open and closed?
 - Yes. For example $\{\}$ and $\mathbb{R}$ are both open and closed.

Bounded Set

- A set $U \subset X$ is *bounded* if $\exists r > 0$ and $x \in X$ such that $U \subset B(x, r)$
 - A set is bounded if we can fit it into a large enough ball around some point. A set is not bounded if no matter how large we choose the radius of the ball, the set will not be completely contained in it.

Compact Set

- A set U is *compact* if it is closed and bounded. A compact set in $\mathbb{R}^N$ fits into a ball and contains its boundary.

Convex Combination and Convex Set

- Given any finite number of points $\{x_1, \dots, x_n\}$, $x_i \in \mathbb{R}^N$ a point $z \in \mathbb{R}^N$ is a *convex combination* of the points $\{x_1, \dots, x_n\}$ if $\exists \lambda \in \mathbb{R}_+^N$ satisfying $\sum_{i=1}^N \lambda_i = 1$ such that $z = \sum_{i=1}^N \lambda_i x_i$.
 - A convex combination for two points in $\mathbb{R}^2$ form the line segment connecting the two points.
- A set is *convex* if the convex combination of any two points in the set is also contained in the set. If $x_1, x_2 \in X$, then $\alpha x_1 + (1 - \alpha)x_2 \in X$, where $\alpha \in [0, 1]$
 - In $\mathbb{R}^2$ this definition says that a straight line that connects any two points in a set is also contained within the set.

Continuous Functions

The function f is called continuous at the point $p \in X$ if

$$\lim_{x \rightarrow p} f(x) = f(p)$$

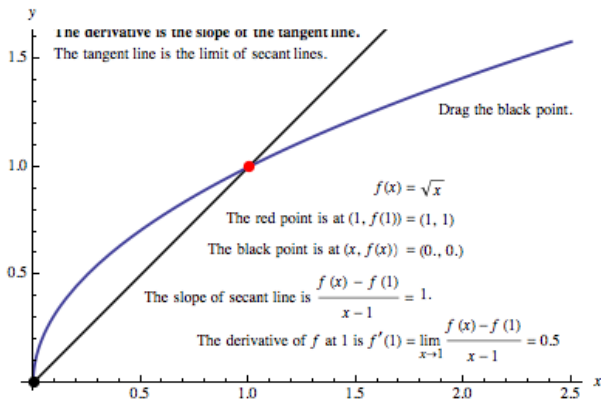
This is equivalent to saying that for every $\epsilon > 0$ we can find $\delta > 0$ such that $|f(x) - f(p)| < \epsilon$ for all x for which $|x - p| < \delta$. f is called continuous if it is continuous at every point of X .

- A continuous function is one without any sudden breaks. You can graph the function without taking your pencil off of the paper.
- A discontinuous function has at least one break
- Continuity is an important property for maximizing functions.

Examples of Discontinuous Functions

- $f(x) = \frac{1}{x}$
- $f(x) = \ln(x)$
- $f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ (x - 2)^2 & \text{if } x \geq 0 \end{cases}$

Secant, Tangent, and the Derivative



Differentiation

The function f is *differentiable at x* if

$$\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists. If f is differentiable for all x in its domain, then we say f is a *differentiable function*. We denote the derivative of f at point x as $\frac{d}{dx}f(x)$ or $f'(x)$.

Notation for the Derivative

- $\frac{d}{dx}f(x)$: the derivative of $f(x)$ with respect to x
- $\frac{dy}{dx}$: the derivative of y with respect to x
- $f'(x)$: f prime x
- $D_x f(x)$: the derivative of $f(x)$ with respect to x

Derivative Example 1

Consider the function $f(x) = 3x$:

- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = \lim_{h \rightarrow 0} \frac{3(x+h)-3x}{h}$
- $\lim_{h \rightarrow 0} \frac{3x+3h-3x}{h} = \lim_{h \rightarrow 0} \frac{3h}{h} = 3$

Derivative Example 2

Consider the function $f(x) = x^2$

- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2-x^2}{h}$
- $\lim_{h \rightarrow 0} \frac{x^2+2xh+h^2-x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh+h^2}{h}$
- $\lim_{h \rightarrow 0} \frac{2xh+h^2}{h} = \lim_{h \rightarrow 0} 2x + h = 2x$

Derivative Example 3

Consider the function $f(x) = x^3 + x - 5$

- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{((x+h)^3 + (x+h) - 5) - (x^3 + x - 5)}{h}$
- $\lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 + x + h - 5 - x^3 - x + 5}{h}$
- $\lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 + h}{h}$
- $\lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 + 1 = 3x^2 + 1$
- Constants cancel. Their derivatives are zero.
- Added terms can be treated separately. x became 1.
- Only the second leading term in the expansion of $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$ survives the limit. We can ignore all other terms in the limit

Non-Differentiable Functions

- A function must be continuous over its domain to be differentiable over this domain
- Continuity is a necessary but not sufficient condition.
- Functions that have kinks are not differentiable in the kinks. For example. $f(x) = |x|$.
- Functions are not differentiable when their tangent line is vertical. In this case the slope is equal to infinity.

Partial Derivatives

- Consider the function $f(x, z)$ and say that we want to know how the value of this function changes with a change in x , while leaving z constant.
- To calculate this, we need to calculate the partial derivative, denoted as $\frac{\partial}{\partial x}f(x, y)$.
- For example, if $f(x, z) = 3z^3 - 3z^2 + \sqrt{z} + x$, then $\frac{\partial}{\partial x}f(x, z) = 1$.
- This derivative is same as the derivative for $f(x) = x$.

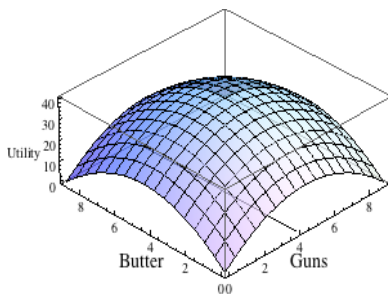
Partial Derivatives

Consider the function $f(x) = f(x_1, x_2, \dots, x_n)$. What is the partial derivative? The partial derivative of f with respect to x_1 , denoted $\frac{\partial f}{\partial x_1}$ or f_{x_1} , is obtained by differentiating f with respect to x_1 and treating all of the other variables as constant:

$$f_{x_1}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1+h, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

Interpretation

$\frac{\partial}{\partial x_1} f(x)$ is the change in the value of the function if x_1 changes infinitesimally. It is the slope of the line tangent to the graph of f at point $\mathbf{x}$ in the x_1 direction. If we have the utility function $u(G, B) = 9G - G^2 + 9B - B^2$, then $\frac{\partial}{\partial B} U(G, B) = -2B + 9$.



Example of Calculating a Partial Derivative

Consider the function $f(x, y) = xy^2$

- $\frac{\partial}{\partial x} f(x, y) = \lim_{h \rightarrow 0} \frac{(x+h)y^2 - xy^2}{h}$
- $\lim_{h \rightarrow 0} \frac{xy^2 + hy^2 - xy^2}{h} = \lim_{h \rightarrow 0} \frac{hy^2}{h} = y^2$

Graphing $f(x, y) = xy^2$

